

```
def find_s(training_data):
    # Student Details
    name = "Zainab Khatoon"
    reg_no = "24BECS173"

    # Initialize the specific hypothesis with the first positive instance
    hypothesis = None

    for row in training_data:
        if row[-1] == 'Yes':
            # Target concept is positive
            hypothesis = row[:-1].copy()
            break

    if hypothesis is None:
        return "No positive instances found."

    # Update hypothesis based on remaining positive instances
    for row in training_data:
        if row[-1] == 'Yes':
            for i in range(len(hypothesis)):
                if row[i] != hypothesis[i]:
                    hypothesis[i] = '?'

    return hypothesis
```

```

1 def find_s(training_data):
5     return hypothesis
6
7
8 # Example Usage
9 if __name__ == "__main__":
10
11     # Student Details
12     name = "Zainab Khatoon"
13     reg_no = "24BECS173"
14
15     # Columns: Sky, Temp, Humidity, Wind, Water, Forecast, EnjoySport
16     dataset = [
17         ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
18         ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
19         ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
20         ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
21     ]
22
23     most_specific_hypothesis = find_s(dataset)
24
25     # Display Student Details
26     print("=====")
27     print("Name      :", name)
28     print("Reg No   :", reg_no)

```

```

# Student Details
name = "Zainab Khatoon"
reg_no = "24BECS173"

# Columns: Sky, Temp, Humidity, Wind, Water, Forecast, EnjoySport
dataset = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

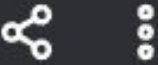
most_specific_hypothesis = find_s(dataset)

# Display Student Details
print("=====")
print("Name      :", name)
print("Reg No   :", reg_no)
print("=====")

print(
    "Most Specific Hypothesis found by Find-S:"

```

Output



=====

Name : Zainab Khatoon

Reg No : 24BECS173

=====

Most Specific Hypothesis found by Find-S: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

=== Code Execution Successful | ===